INTERNAL SPECIFICATION FOR **CERAMIC TILES**

Sr. No	CHARACTERISTICS	SF / Rectified 30x60x0.9 cm		Roofing Tile 32.5X32.5x0.9 cm	
1	Nominal size **(mm)**	300.0x600.0x9.0		325.0x325.0x9.0	
2	Work size **(mm)**	300.5x600.5x9.0		325.5x325.5X9.0	
		ISO 13006	**SCC. spec.**	**ISO 13006**	**SCC. spec.**
3.1	Size variation from work size **(%)**	± 0.30	± 0.30	± 0.60	± 0.60
	In **mm**	± 1.90 ± 1.00	± 0.50 ± 0.50	± 1.95	± 1.95
3.2	Size range in **mm**	299.6 – 301.4 599.5 – 601.5	299.6 – 300.1 599.6 - 600.1	323.5 - 327.5	323.5 - 327.5
3.3	Size code 03	-	-	-	323.5 – 324.0
	Size code 01	-	-	-	324.1 – 325.0
	Size code 00 **(Normal)**	-	299.6 – 300.1 599.6 - 600.1	-	325.1 – 326.0
	Size code 02	-	-	-	326.1 – 327.0
	Size code 04	-	-	-	327.0 – 327.5
4	Thickness deviation **(%)**	± 10.0	± 5.0	± 10.0	± 10.0
5	Thickness Range **(mm)**		8.55 – 9.45		8.50 - 9.50
6	Centre Curvature	mm	mm	mm	mm
6.1	a) Convex **(+)**	+ 1.80	+1.50	+ 2.00	+ 2.00
6.2	b) Concave **(-)**	-1.80	-0.30	- 2.00	- 2.00
7	Warpage	± 1.80	± 1.30	± 2.00	± 2.00
8	Edge Curvature				
8.1	a) Convex **(+)**	+ 1.80	+1.30	+ 1.63	+ 1.63
8.2	b) Concave **(-)**	-1.80	-0.40	- 1.63	- 1.63
9	Max deviation from Straightness (**mm)**	± 0.80	± 0.50	± 1.50	± 1.50
10	Max deviation from Rectangularity (**mm)**	± 1.50	± 1.20	± 1.59	± 1.63
11	Water Absorption **(%)**	> 10	16 ± 2	> 10	16 ± 2
12	Coefficient to Linear thermal expansion **($x10^{-6}K^{-1}$)**	-	≤ 8	-	≤ 8
13	Resistance to surface abrasion **(Class)**	Manufacturer to state class	2 - 5	Manufacturer to state class	2 - 5
14	Breaking strength **(N)**	≥ 600	≥ 800	≥ 600	≥ 600
15	Modulus of Rupture **(N/mm^2)**	Min. 12	Min 15	Min. 12	Min 12
16	Crazing resistance **(5bar/2 hr cycle)**	Required	Resistant	Required	Resistant
17	Scratch resistance **(Moh's Scale)**	-	Min.3 (Glossy) Min.6 (Matt)	-	Min.3 (Glossy) Min.6 (Matt)
18	Staining resistance **(Min.Class)**	3	3	3	3
19	Chemical resistance to Acids and alkalis **(Min.Class)**	Required if agreed	B	Required if agreed	B
20	Resistance to household Chemicals **(Min.Class)**	B	B	B	B